

A NOTE ON LIKELIHOOD ESTIMATION OF A SIMPLE REGRESSION MODEL WITH MEASUREMENT ERROR VIA THE ORTHOGONAL PARAMETRIZATION

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Summary: In this paper, working with an orthogonal parametrization, conditional likelihood based methods for obtaining exact and approximate confidence intervals for the slope in a simple linear regression model with measurement error are discussed. Our setup is simple and different from the one considered by Wong (1989).

1. INTRODUCTION

The classical simple linear regression analysis assumes that the independent variable is defined by

$$(1) \quad Y_t = \alpha + \beta x_t + e_t, \quad t = 1, \dots, n,$$

where $(x_1, \dots, x_n)$ is fixed in repeated sampling and e_t are independent $N(0, \sigma_{ee}^2)$ random variables. It is assumed that x_t is measured without error. However, in practice, particularly in social sciences and biological assay, this assumption is often violated. There is a lot of work on the problem of parameter estimation when the x_t contain errors of measure-

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ment, see for example, Madansky (1959), Sprent (1966), Kendall & Stuart (1961) and recently Fuller (1987). In the present paper we try to solve the estimation problem by working with a conditional orthogonal likelihood model, when only the independent variable is measured with error. We shall study models of type (1), with $\alpha = 0$, where instead of observing x_i one observes the sum

$$(2) \quad X_i = x_i + u_i.$$

We make the customary assumption that

$$(3) \quad (x_i, u_i, e_i) \sim N((\mu_x, 0, 0), \text{diag}(\sigma_{xx}^2, \sigma_{uu}^2, \sigma_{ee}^2)).$$

We can without loss of generality take $\mu_x = 0$. Because (X_i, Y_i) is distributed as a bivariate normal, the conditional distribution of Y_i given X_i is normal with mean θX_i and variance $\sigma_{uu}^2 K_{xx} \beta^2 + \sigma_{ee}^2$, where

$$(4) \quad K_{xx} = \frac{\sigma_{xx}^2}{\sigma_{xx}^2 + \sigma_{uu}^2} \quad \text{and}$$

$$(5) \quad \theta = K_{xx} \beta.$$

From (5) we see the effect of the measurement error, that is, the regression coefficient β has been attenuated by the factor K_{xx} . The ratio K_{xx} is called the reliability of X_i in the social science. Therefore as suggested by Fuller (1987), we call K_{xx} the reliability ratio. Working with an orthogonal parametrization (Cox and Reid, 1987), we constructed exact and approximate confidence intervals for β when the reliability ratio is known and unknown, respectively.

2. EXACT CONFIDENCE INTERVAL WITH KNOWN RELIABILITY RATIO

In this section, our main interest is to make inference about β under the assumption that the reliability ratio is known. Assuming the x is a particular trait of the individual, there are a number of situations, particularly in psychology and sociology, where the factor K_{xx} is known. The IQ test

is the best known example. The value of the trait x for a particular individual is impossible to observe directly, and the response X_t is obtained as a score constructed from the answers to a number of questions. When the questions have been studied, it is possible to get an idea of K_{xx} . In general, the factor is published together with the supporting material. The procedure to obtain a confidence interval for β is simplified by working with the orthogonal parametrization,

$$(6) \quad \lambda = \sigma_{uu}^2 K_{xx} \beta^2 + \sigma_{ee}^2.$$

If $(X_1, Y_1), \dots, (X_n, Y_n)$ are independently and identically distributed according to model (3), the conditional likelihood function in the new parametrization is

$$(7) \quad \ell(\beta, \lambda) = -\frac{n}{2} \ell n(2\pi) - \frac{n}{2} \ell n \lambda - \frac{1}{2\lambda} \sum_{t=1}^n (Y_t - \beta K_{xx} X_t)^2.$$

It is well known that maximum likelihood estimators of β and λ are

$$\begin{aligned} \hat{\beta} &= \frac{\sum_{t=1}^n K_{xx} X_t Y_t}{\sum_{t=1}^n K_{xx}^2 X_t^2} \\ &= K_{xx}^{-1} \frac{\sum_{t=1}^n X_t Y_t}{\sum_{t=1}^n X_t^2} \\ (8) \quad &= K_{xx}^{-1} \hat{\beta}_1 \quad \text{and} \\ (9) \quad \hat{\lambda} &= \frac{1}{n-1} \sum_{t=1}^n (Y_t - \hat{\beta}_1 X_t)^2. \end{aligned}$$

The same result was obtained by Fuller (1987) using a different approach and a more general model. Note the simplicity in obtaining the results here by working with the orthogonal parameters. The exact $1 - \alpha$ level confidence interval for β formed via the studentized deviate is

$$(10) \quad \left[\hat{\beta} - t_{\frac{1}{2}\alpha} \sqrt{\frac{\hat{\lambda}}{K_{xx}^2 \sum_{t=1}^n X_t^2}}, \hat{\beta} + t_{\frac{1}{2}\alpha} \sqrt{\frac{\hat{\lambda}}{K_{xx}^2 \sum_{t=1}^n X_t^2}} \right].$$

For testing the hypothesis $\beta = \beta_0$, we can use the t-statistic

$$(11) \quad t = \frac{\hat{\beta} - \beta_0}{\sqrt{\frac{\hat{\lambda}}{K_{xx} \sum_{i=1}^n X_i^2}}} \sim t(n-1).$$

3. APPROXIMATE CONFIDENCE INTERVAL WITH UNKNOWN RELIABILITY RATIO

It is well known that, when both σ_{uu}^2 and σ_{ee}^2 are unknown, β is not identifiable. Model (1)-(2), with the known ratio $\delta = \sigma_{ee}^2/\sigma_{uu}^2$, is called the error-in-variables model. See Fuller's book (1987) p.30 for interesting comments on that model. In this section, we consider the "overidentifiable" situation, that is, $\sigma_{ee}^2 = \sigma_{uu}^2 = \sigma_0^2$, where the common variance is known. The "overidentifiable case" has aroused considerable comments and the maximum likelihood estimates were obtained by Barnett (1967). In this special case, we consider the parameter of interest as $\varphi = \theta$ and the nuisance parameter as $\phi = \beta$. Using the argument of Section 3, we have from (6) that

$$(12) \quad \lambda = \sigma_0^2(\phi\varphi + 1).$$

It is easy to see that λ is orthogonal to φ and that its natural estimators are

$$(13) \quad \begin{aligned} \hat{\theta} &= \hat{\beta}_1 \\ \hat{\lambda} &= \frac{\sum_{i=1}^n (Y_i - \hat{\beta}_1 X_i)^2}{n-1}. \end{aligned}$$

$$(14) \quad \sqrt{n} \begin{pmatrix} \hat{\theta} - \theta \\ \hat{\lambda} - \lambda \end{pmatrix} \Rightarrow N \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \frac{\lambda}{\sigma_{xx}^2} & 0 \\ 0 & 2\lambda^2 \end{pmatrix} \right),$$

where $\sigma_{xx}^2 = \sigma_{xx}^2 + \sigma_{uu}^2$. Since we are interested to find an approximate confidence interval for β , we consider, from (12), the following reparametrization:

$$(15) \quad \beta = \frac{\frac{\lambda}{\sigma_0^2} - 1}{\theta}$$

From (14)-(15) we obtain

$$(16) \quad \sqrt{n}(\hat{\beta} - \beta) \Rightarrow N(0; \sigma^2), \text{ where}$$

$$\sigma^2 = \left(\frac{\frac{\lambda}{\sigma_0^2} - 1}{\theta^2} \right)^2 \frac{\lambda}{\sigma_{XX}^2} + 2\lambda^2 \frac{1}{\sigma_0^2 \theta^2}, \quad \text{and}$$

$$(17) \quad \hat{\beta} = \frac{\frac{\lambda}{\sigma_0^2} - 1}{\hat{\theta}}.$$

An approximate $(1 - \alpha)$ -level confidence interval for β formed via the studentized deviate, is

$$(18) \quad \left(\hat{\beta} - z_{\frac{1}{2}\alpha} \sqrt{\frac{\widehat{\sigma^2}}{n}}, \hat{\beta} + z_{\frac{1}{2}\alpha} \sqrt{\frac{\widehat{\sigma^2}}{n}} \right)$$

where $\widehat{\sigma^2}$ and $\hat{\beta}$ are obtained by (13) from (15) and (16), with $\widehat{\sigma_{XX}^2} = \sum_{i=1}^n X_i^2/n$. It is interesting to note that if we take $\varphi = \beta$ and $\phi = (K_{xx}, \beta)$, as pointed out by Cox and Reid (1987), the orthogonal parametrization cannot be obtained analytically.

4. SIMULATION STUDY

In this section, we illustrate the performance of our procedure based on a simulation of size 500. The simulation was carried out on a PC/XT micro-computer using the Minitab program to generate pseudo-random numbers from the appropriate normal distribution.

Table 1 shows the good performance of the coverage frequencies of the approximate intervals (K_{xx} unknown). Table 2 shows that the decrease in length of the exact intervals is considerable for $\sigma_{xx} = 8.0$. It is interesting to note that the increase in length of the confidence intervals of Wong (1989), for $\sigma_{xx} = 1$, is greater than our approximate intervals for small n . Also, Table 2 shows that the performance of the exact intervals is better than the approximate intervals, as expected.

Table 1

Five hundred sets of simulated data for $\alpha = 0, \beta = 1, \sigma_0^2 = 1$ for different σ_{xx} and n . Comparison of confidence intervals of β for different coverages when K_{xx} is unknown.

σ_{xx}	n	0.95	0.90
		K_{xx} unknown	K_{xx} unknown
1.0	50	0.91	0.90
1.0	30	0.90	0.87
1.0	20	0.89	0.84
4.0	50	0.92	0.88
4.0	30	0.90	0.87
4.0	20	0.89	0.87
8.0	50	0.94	0.87
8.0	30	0.91	0.88
8.0	20	0.92	0.84

Table 2

Five hundred sets of simulated data for $\alpha = 0, \beta = 1, \sigma_0^2 = 1$ and different σ_{xx} and n . Comparison of average widths of confidence intervals with coverage 0.95 when K_{xx} is known and unknown.

n	$\sigma_{xx} = 1.0$		$\sigma_{xx} = 4.0$		$\sigma_{xx} = 8.0$	
	K_{xx} unknown	K_{xx} known	K_{xx} unknown	K_{xx} known	K_{xx} unknown	K_{xx} known
20	6.988	1.671	2.648	0.401	2.553	0.189
30	5.722	1.328	2.129	0.320	2.026	0.150
50	2.946	1.002	1.682	0.243	1.586	0.114

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